DOCUMENT C | SOLUTIONS 1

Worked solutions and proposed marking scheme

Best-fit predicted practice paper. These are independent suggested answers and mark allocations, not official university solutions.

Award method marks for correct reasoning and units even when a later arithmetic slip occurs; avoid penalizing the same carried error repeatedly. Equivalent physically correct explanations and valid stated assumptions can earn credit. A part's marking points total its question allocation. Diagrams are schematic and not to scale. The complete question text is repeated below so each solution file is usable independently.

Source notation

S1-S06 refer to the supplied six lecture decks and PDF page/slide numbers. A1/A2 are the supplied thermal/OCT student assignments, cited by PDF page including covers. "Recurring family" identifies the historical concept, not copied wording; original extensions are labeled. A historical reference means the closest related question, not evidence that the exact model task appeared.

Equation corrections

Positive activity is -dN/dt. Conventional spin echoes use intrinsic T2 in the ideal echo envelope, whereas gradient echoes retain T2* sensitivity. RF bandwidth means full width unless explicitly stated otherwise. All calculations use γ=267.52×10^6 rad s^-1 T^-1 and unrounded intermediate values.

Model 1 | Question 1: Image quality and nuclear medicine

1(a) [3 marks]

Question. Define spatial resolution. State what the point spread function (PSF) represents and write the image-formation relationship for a linear, shift-invariant system.

Expected answer and working. Spatial resolution is the minimum separation at which distinct small features can be resolved. The PSF h is the image of an ideal point object. For a linear, shift-invariant system, I(x,y)=O(x,y)*h(x,y), with * denoting convolution. A finite-width PSF spreads object signal over neighbouring positions.

Proposed allocation: Resolution definition (1); Point response (1); Convolution with stated assumption (1). **Total 3.**

Source map: S1 pp.39,45-48. **Closest past question:** 2025 Q1(a)i-iii. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

1(b) [4 marks]

Question. A line spread function is L(x)=C exp[-(x-x0)^2/(2σ^2)]. Derive its full width at half maximum (FWHM). Evaluate the width for σ=0.60 mm.

Expected answer and working. At the peak L(x0)=C. At each half-height point, exp[-(x-x0)^2/(2σ²)]=1/2. Taking natural logs gives (x-x0)²=2σ² ln2, so the two points are x0±σ√(2 ln2). Their separation is FWHM=2√(2 ln2)σ=2.35482σ. For σ=0.60 mm, FWHM=1.41289 mm, approximately 1.41 mm.

Proposed allocation: Half-height equation (1); Logarithm and two roots (1); Full-width expression (1); 1.41 mm with unit (1). **Total 4.**

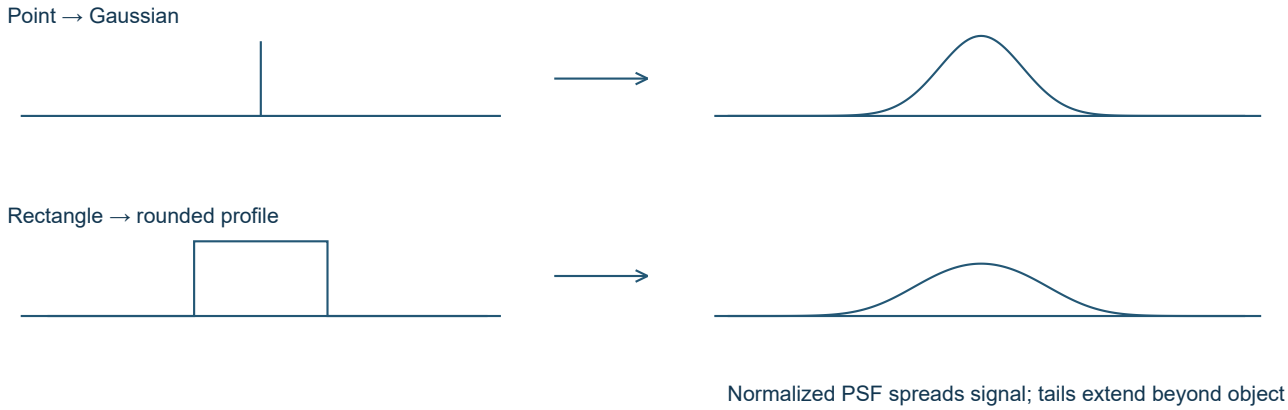
Source map: S1 pp.41-44. **Closest past question:** 2024 Q1(a)iii. **Status:** Recurring family.

Typical mark-losing error: Do not report the half-width σ√(2 ln2) as FWHM.

1(c) [3 marks]

Question. Sketch the result of convolving (i) an ideal point and (ii) a narrow rectangular object with a normalized Gaussian PSF. Explain the effect on fine structures.

Expected answer and working. An ideal point becomes the Gaussian PSF (scaled by its point strength). The rectangle has rounded edges, nonzero tails and a reduced peak when it is narrow relative to the PSF. Convolution preserves integrated signal for a unit-area PSF, but spreads it; closely spaced fine features can merge.



Proposed allocation: Point becomes Gaussian (1); Rounded rectangle and tails (1); Loss of fine detail / merging (1). **Total 3.**

Source map: S1 pp.45-48. **Closest past question:** 2025 Q1(a)ii-iii. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

1(d) [4 marks]

Question. State four desirable properties of a SPECT radiotracer, explaining the imaging reason for each.

Expected answer and working. Choose a physical half-life long enough for distribution and acquisition but short enough to limit unnecessary exposure; predominantly monoenergetic gamma emission with little unnecessary particulate dose; gamma energy suitable for tissue escape and collimator/detector performance (the lecture suggests roughly 100-200 keV); and high target uptake with low background uptake to improve contrast.

Proposed allocation: Suitable half-life + reason (1); Gamma emission + reason (1); Suitable energy + reason (1); Target/background specificity + reason (1). **Total 4.**

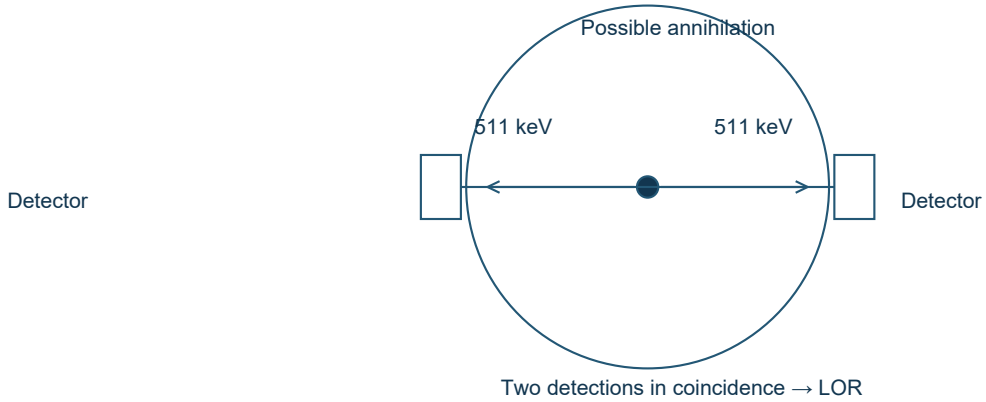
Source map: S06 pp.9-10. **Closest past question:** 2024 Q1(b)i. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

1(e) [5 marks]

Question. Using a labeled sketch, explain positron annihilation coincidence detection in PET. Give two benefits of combining PET with CT.

Expected answer and working. The tracer emits a positron which slows in tissue and annihilates with an electron. The dominant two-photon event produces two approximately opposite 511 keV photons. Opposing detector signals accepted within a coincidence time window define a line of response (LOR), rather than an exact point. Many LORs reconstruct tracer distribution. CT supplies anatomical localization and an attenuation map for PET correction.



Proposed allocation: Positron annihilation and 511 keV pair (1); Opposing detectors + coincidence window (1); LOR sketch / reconstruction (1); Anatomical localization (1); Attenuation correction (1). **Total 5.**

Source map: S06 pp.25-34. **Closest past question:** 2025 Q1(b)ii-iii. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

1(f) [6 marks]

Question. At t=0, two patients have in-body activities AA=240 MBq and AB=80 MBq. A has physical and biological half-lives of 6 h and 6 h. B has physical and biological half-lives of 12 h and 12 h. Assuming independent exponential clearance, find when their in-body activities are equal and the common activity.

Expected answer and working. Independent physical decay and clearance add their rate constants: 1/Teff=1/Tphys+1/Tbio. Thus Teff,A=3 h and Teff,B=6 h. Set 240·2^(-t/3)=80·2^(-t/6). Dividing gives 3=2^(t/6), hence t=6 ln3/ln2=9.50978 h. Substitution into B gives AB=80/3=26.6667 MBq. A starts higher but decreases faster; the equality is a single positive crossing.

Proposed allocation: Effective-rate law (1); Both effective half-lives (1); Equality equation with initial ratio (1); Log solution (1); 9.51 h (1); 26.7 MBq and interpretation (1). **Total 6.**

Source map: S06 pp.6-8. **Closest past question:** 2024 Q1(b)iv. **Status:** Recurring family.

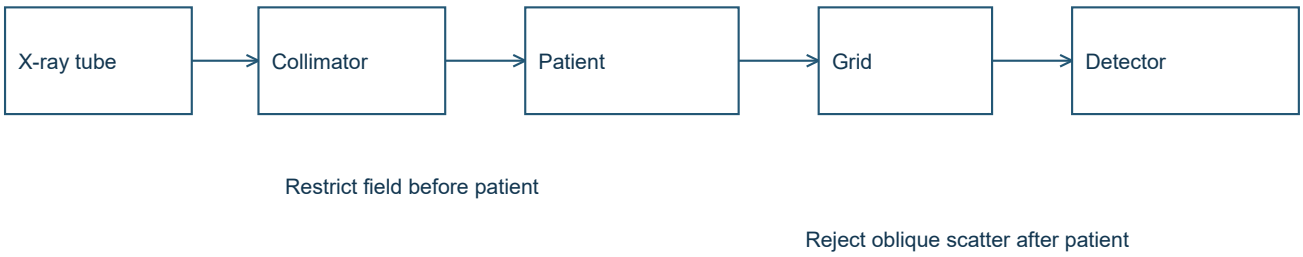
Typical mark-losing error: Use effective half-lives for in-body activity; do not add the half-lives themselves.

Model 1 | Question 2: Projection radiography and computed tomography

2(a) [4 marks]

Question. Draw a planar X-ray radiography system, labeling the tube, beam collimator, patient, anti-scatter grid and detector. State the distinct functions of collimator and grid.

Expected answer and working. The tube produces X-rays; the upstream collimator restricts the irradiated field before the patient. The downstream grid preferentially absorbs obliquely scattered photons before they reach the detector. The detector converts transmitted radiation to a recordable image. Collimation reduces unnecessary exposure and scatter generation; the grid reduces detected scatter but also absorbs some useful primary photons.



Proposed allocation: Correct source-to-detector order (1); Tube and detector functions (1); Collimator restricts field (1); Grid rejects scatter (1). **Total 4.**

Source map: S2 pp.32-42. **Closest past question:** 2025 Q2(a)i. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

2(b) [3 marks]

Question. State three consequences or trade-offs of reducing the effective focal spot. Include an image-quality consequence.

Expected answer and working. A smaller effective spot reduces geometric penumbra and improves spatial resolution. Achieving it by decreasing the anode angle reduces useful beam coverage. Achieving it by concentrating electron power into a smaller actual target area constrains tube loading because of local heating. Credit comparable physically justified consequences; distinguish actual and projected spot size.

Proposed allocation: Less geometric blur / improved detail (1); Coverage trade-off with anode angle (1); Heat or permissible loading trade-off (1). **Total 3.**

Source map: S2 pp.7-13,50. **Closest past question:** 2025 Q2(a)ii-iii. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

2(c) [5 marks]

Question. A monoenergetic beam illuminates 0.60 cm thick bars with HVL=0.20 cm. Gaps have negligible attenuation. Calculate μ, Ibar/Igap, and Weber contrast C=(Igap-Ibar)/Igap. State one reason this simple model is imperfect for a clinical polychromatic beam.

Expected answer and working. HVL=ln2/μ gives μ=0.693147/0.20=3.46574 cm^-1. There are three HVLs in 0.60 cm, so Ibar/Igap=exp(-μ·0.60)=2^-3=0.125. The specified Weber contrast is 1-0.125=0.875 (87.5%). A clinical beam has a spectrum: lower energies are preferentially removed, causing beam hardening and a thickness-dependent effective attenuation; scatter is another valid limitation.

Proposed allocation: HVL relationship (1); μ=3.47 cm^-1 (1); Transmission 0.125 (1); Contrast 0.875 (1); Physical limitation (1). **Total 5.**

Source map: S2 pp.27-31; S3 p.59. **Closest past question:** 2024 Q2(a)iii. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

2(d) [3 marks]

Question. Give three advantages of CT over planar projection radiography.

Expected answer and working. CT separates anatomy into cross-sectional slices, reducing superposition; it has better ability to distinguish small attenuation differences in soft tissue; and calibrated CT numbers permit quantitative attenuation comparisons and geometrical measurements. These advantages coexist with higher typical radiation exposure and reconstruction requirements.

Proposed allocation: Reduced superposition (1); Improved low-contrast discrimination (1); Quantitative HU or accurate 3D localization (1). **Total 3.**

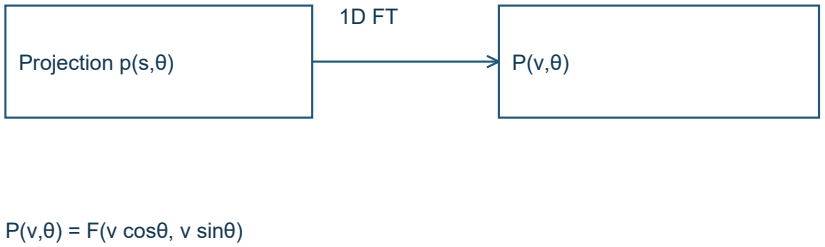
Source map: S3 pp.2-3,31-32. **Closest past question:** 2024 Q2(b)i. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

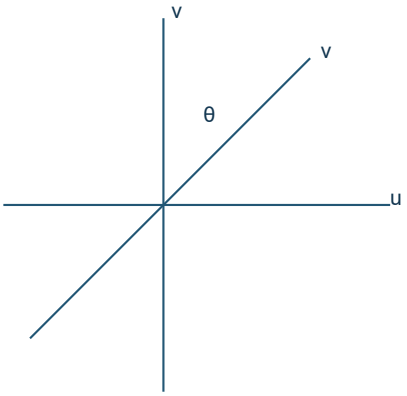
2(e) [4 marks]

Question. State the central section theorem, with a labeled diagram relating projection space to the object Fourier domain.

Expected answer and working. For $p(s,\theta)=\int f(s\cos\theta-t\sin\theta, s\sin\theta+t\cos\theta)dt$, take the 1D FT in s. It equals the 2D FT $F(u,v)$ of the object sampled on the line $(u,v)=(v\cos\theta, v\sin\theta)$ through the origin. The same projection-normal angle θ fixes the radial line. Thus projections at many angles fill radial Fourier samples, followed by interpolation and inverse transformation.



$P(v,\theta) = F(v\cos\theta, v\sin\theta)$



Radial slice of F(u,v)

Proposed allocation: 1D FT of projection (1); Equals radial slice of 2D object FT (1); Same angle / through origin (1); Correct labeled diagram (1). **Total 4.**

Source map: S3 pp.15-16. **Closest past question:** 2024 Q2(b)ii. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

2(f) [6 marks]

Question. A 2×2 image has unit-weight ray sums: top row 6, bottom row 10, left column 7 and right column 9. Start all pixels at zero. Perform one ART sweep in that order, adding each ray residual equally to its two pixels. Report every intermediate image and determine whether these data uniquely identify all four pixels.

Expected answer and working. Let the image be [[a,b],[c,d]]. After top row: [[3,3],[0,0]]. After bottom row: [[3,3],[5,5]]. The predicted left sum is 8; residual 7-8=-1, so add -0.5 to its two pixels: [[2.5,3],[4.5,5]]. The right sum is 8; residual 9-8=1, so add +0.5: [[2.5,3.5],[4.5,5.5]]. All four sums now match. They do not imply uniqueness: [[a,6-a],[7-a,3+a]] has the same sums. More independent views or constraints are needed to determine the actual image.

Proposed allocation: Two row updates (2); Left residual/update (1); Right residual/final image (1); Verification of sums (1); Nonuniqueness with reason or family (1). **Total 6.**

Source map: S3 pp.26-28. **Closest past question:** 2024 Q2(b)iii. **Status:** Recurring family.

Typical mark-losing error: A fit to a few projections is not proof of a unique reconstruction.

Model 1 | Question 3: Magnetic resonance imaging

3(a) [3 marks]

Question. Name three intrinsic tissue parameters that affect MRI pixel intensity and explain why intensity cannot be interpreted as proton density alone.

Expected answer and working. Proton density ρ, longitudinal relaxation time T1 and transverse relaxation time T2 (or T2* in GE) affect signal. The acquisition TR, TE and flip angle weight their contributions. Two tissues with equal proton density can therefore have different brightness because they recover or dephase at different rates.

Proposed allocation: ρ (1); T1 (1); T2/T2* with sequence dependence (1). **Total 3.**

Source map: S4 pp.3,75,78-79. **Closest past question:** 2024 Q3(a)i; 2025 Q3(a)i. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

3(b) [4 marks]

Question. A 1.5 T scanner uses Gz=20 mT/m to select a 4.0 mm thick axial slice centred at z=0. Calculate the central Larmor frequency, full RF excitation bandwidth and frequency endpoints in Hz.

Expected answer and working. Use γbar=γ/(2π)=42.5771304 MHz/T. Then f0=γbar·1.5=63.8656956 MHz. Convert Gz=0.020 T/m and Δz=0.0040 m. Full Δf=γbar·Gz·Δz=3406.1704 Hz. Endpoints are f0±Δf/2: 63,863,992.48 Hz and 63,867,398.65 Hz (approximately 63.86399-63.86740 MHz). The bandwidth spans the full slab, not just one side.

Proposed allocation: Hz conversion / f0 (1); Correct G and thickness units (1); Full BW=3406 Hz (1); Endpoints f0±1703 Hz (1). **Total 4.**

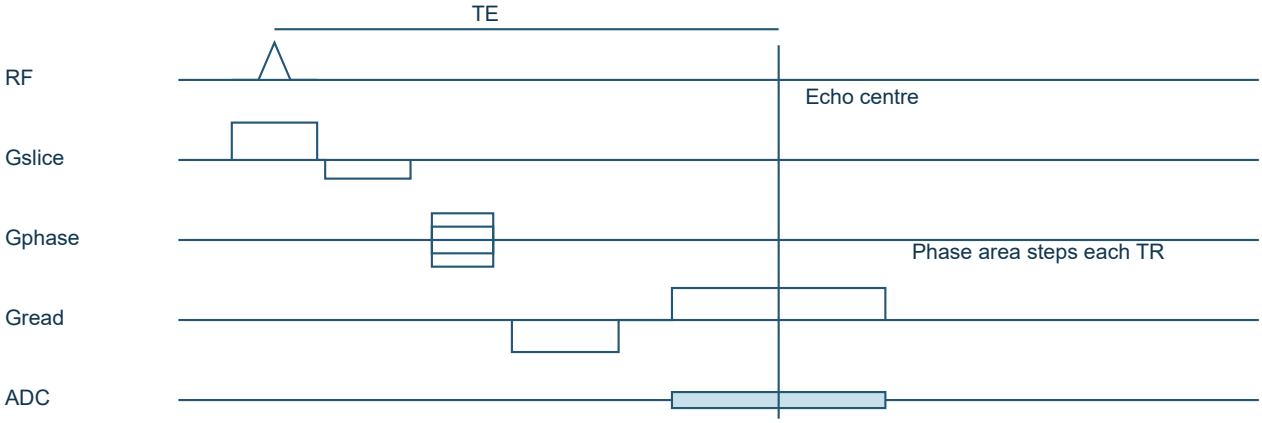
Source map: S4 pp.15,48-54. **Closest past question:** 2023 Q3(b)ii-iii; 2025 Q3(a)ii. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

3(c) [5 marks]

Question. Sketch a basic gradient-echo imaging sequence on RF, Gslice, Gphase, Gread and ADC lines. Identify TE and explain the purpose of each gradient.

Expected answer and working. Apply frequency-selective RF concurrently with a slice gradient, then a slice-rephasing lobe. Apply a phase-encoding lobe whose area changes between TR repetitions to select ky. A readout prephaser dephases spins before the opposite-polarity readout gradient produces the echo; acquire data under the readout gradient around its echo centre. TE is RF excitation centre to echo centre. No 180-degree RF refocusing pulse is used in basic GE.



Proposed allocation: RF simultaneous with slice selection (1); Slice rephaser (1); Variable phase-encoding lobe (1); Read prephaser/readout and ADC (1); TE correctly labeled and gradient purposes (1). **Total 5.**

Source map: S4 pp.52-65. **Closest past question:** 2024 Q3(b)i; 2025 Q3(b)i. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

3(d) [4 marks]

Question. For one average, 192 phase-encoding steps and TR=0.80 s, calculate acquisition time. Each slice occupies a complete 20 ms sequence slot. Ignoring overhead and cross-talk, find how many slices can be interleaved without increasing that time. Explain how the selected slice is changed.

Expected answer and working. One phase-encoding line is acquired per TR for each slice. Tscan=Nphase·TR·NEX=192·0.80·1=153.6 s (2 min 33.6 s). A complete 20 ms slot gives floor(800/20)=40 slice slots per TR. Change RF centre frequency under the same slice-selection gradient to excite a different z position; repeat the same phase step for each slice and store separately. The stated full slot, not merely TE, sets this ideal packing bound.

Proposed allocation: 153.6 s (1); 40 slice slots (1); RF centre frequency changes slice position (1); Each slice retains 0.80 s TR and separate data (1). **Total 4.**

Source map: S4 pp.61-74. **Closest past question:** 2025 Q3(b)iii-iv. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

3(e) [4 marks]

Question. A spoiled steady-state GE scan has TR=0.50 T1. Find the Ernst angle. For fixed TR, TE and receiver noise, calculate the percentage SNR reduction when a 90° pulse replaces the optimum angle.

Expected answer and working. E1=exp(-TR/T1)=exp(-0.5)=0.60653066. The optimum is αE=acos(E1)=52.66093°. The common ρ·exp(-TE/T2*) factor cancels. At 90°, S90=1-E1=0.39346934. At αE, SE=(1-E1)√(1-E1²)/(1-E1²)=√[(1-E1)/(1+E1)]=0.49489258. Thus S90/SE=√(1-E1²)=0.79506010. With the same noise, SNR loss=100(1-0.79506010)=20.49399%, approximately 20.5%.

Proposed allocation: E1 and optimum angle (1); Correct two signals / ratio setup (1); Ratio 0.7951 (1); Loss 20.5% (1). **Total 4.**

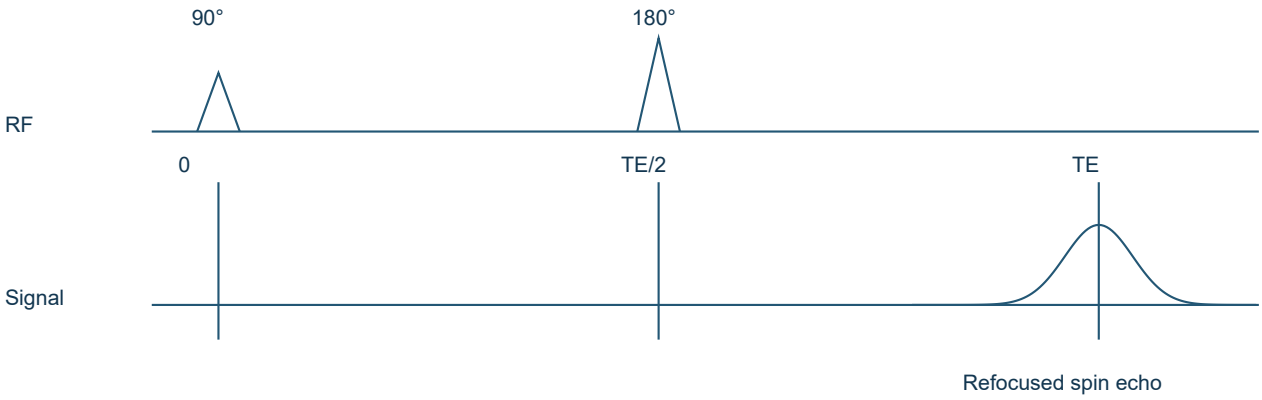
Source map: S4 p.78. **Closest past question:** 2024 Q3(b)iv-v. **Status:** Recurring family.

Typical mark-losing error: Compare signals at identical TR/TE/noise; sin(α) alone omits steady-state saturation.

3(f) [5 marks]

Question. Explain why a conventional spin echo is less sensitive than GE to static magnetic-field inhomogeneity. Sketch the RF and signal timing and state which transverse relaxation time governs the ideal echo envelope.

Expected answer and working. A 90° excitation creates transverse magnetization; static field offsets make spins accumulate different phases. A 180° RF pulse at TE/2 reverses their relative phase positions. Stationary spins with constant offsets rephase at TE, producing an echo. Intrinsic irreversible T2 decay is not recovered. The ideal spin-echo envelope is exp(-TE/T2); gradient echo alone does not reverse static field-offset dephasing and retains T2* effects.



Proposed allocation: Static offsets cause dephasing (1); 180° at TE/2 (1); Rephasing at TE (1); Irreversible T2 remains / SE envelope (1); Labeled RF and signal sketch / GE comparison (1). **Total 5.**

Source map: S4 pp.79-81. **Closest past question:** 2025 Q3(b)v-vi. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

Model 1 | Question 4: Ultrasound and optical coherence tomography

4(a) [3 marks]

Question. Explain why an air gap between the ultrasound probe and skin is undesirable and how coupling gel helps.

Expected answer and working. Acoustic impedance $Z=\rho c$ differs greatly between tissue and air. The intensity reflection coefficient at normal incidence is $[(Z2-Z1)/(Z2+Z1)]^2$, close to unity for tissue/air, so little energy enters the body. Gel replaces the air gap with an acoustic coupling layer and improves transmission.

Proposed allocation: Large impedance mismatch (1); Strong reflection / weak transmission (1); Gel displaces air (1). **Total 3.**

Source map: S05 pp.11-15. **Closest past question:** 2025 Q4(a)i. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

4(b) [3 marks]

Question. Explain ultrasound speckle and give two acquisition methods for reducing it.

Expected answer and working. Speckle is the granular pattern caused by constructive and destructive interference of echoes from unresolved scatterers. Angular compounding combines views obtained with different beam directions; frequency compounding combines sufficiently different frequencies. The true structures remain aligned while speckle is partly decorrelated. Repeating an identical view does not give the same improvement as averaging independent electronic noise.

Proposed allocation: Coherent interference mechanism (1); Angular compounding (1); Frequency compounding (1). **Total 3.**

Source map: S05 pp.32-35. **Closest past question:** 2024 Q4(a)ii-iii; 2025 Q4(a)ii. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

4(c) [5 marks]

Question. A small target lies 15 cm deep. Tissue attenuation is 0.80 dB/(cm·MHz). At most 80 dB propagation loss is allowed. Choose among 2, 3 and 5 MHz, ignoring reflection losses. Calculate the loss at each frequency and justify the choice for best resolution. Use $c=1540$ m/s.

Expected answer and working. The pulse travels 30 cm on its round trip. $L=2\cdot0.80\cdot f\cdot15=24f$ dB, so losses are 48, 72 and 120 dB at 2, 3 and 5 MHz. Only 2 and 3 MHz meet the 80 dB budget. Choose 3 MHz: for a fixed number of cycles per pulse, axial resolution improves as 1/f, and focal lateral resolution also improves with smaller wavelength. At 3 MHz $\lambda=1540/(3e6)=0.5133$ mm; 5 MHz is excluded by the loss budget.

Proposed allocation: Round-trip distance (1); Three losses (2); 3 MHz with threshold reasoning (1); Resolution justification (1). **Total 5.**

Source map: S05 pp.16,26-29. **Closest past question:** 2023 Q4(a)iii; 2025 Q4(a)iii-iv. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

4(d) [4 marks]

Question. A B-mode array has 192 elements fired in non-overlapping groups of six, one line per group. At 15 cm depth and c=1540 m/s, find the scan lines per frame and maximum frame rate. Assume one pulse per line, sequential lines and no overhead.

Expected answer and working. There are 192/6=32 independent group firings and hence 32 scan lines. Deepest echo time is 2d/c=0.30/1540=194.805 µs. Frame time is 32·194.805 µs=6.23377 ms; maximum frame rate=160.4167 frames/s. Real timing overhead or extra focal-zone transmissions lowers this bound.

Proposed allocation: 32 lines (1); Round-trip time (1); Frame time or correct formula (1); 160.4 frames/s with upper-bound interpretation (1). **Total 4.**

Source map: S05 pp.6-7,17,19,23-24. **Closest past question:** 2024 Q4(a)iv-v. **Status:** Recurring family.

Typical mark-losing error: The number of image rows or depth pixels does not multiply the pulse travel time.

4(e) [4 marks]

Question. Describe the physical cause and image appearance of (i) reverberation and (ii) posterior acoustic enhancement.

Expected answer and working. Reverberation occurs when a pulse reflects repeatedly between strong interfaces (often probe and a nearby strong reflector). Longer travel paths are mapped as deeper echoes, producing repeated, often equally spaced lines. Enhancement occurs behind a region that attenuates less than the surrounding tissue, such as a fluid-filled region; deeper echoes are brighter than expected under the same gain compensation.

Proposed allocation: Reverberation mechanism (1); Repeated-line appearance (1); Low-attenuation path mechanism (1); Brighter posterior region (1). **Total 4.**

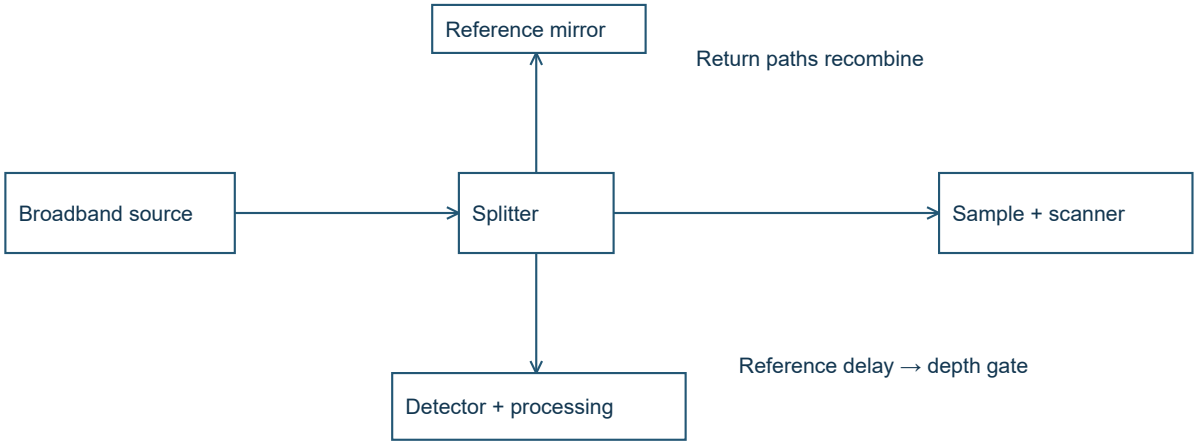
Source map: S05 pp.36-37. **Closest past question:** 2024 Q4(a)ii. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

4(f) [6 marks]

Question. Draw a basic time-domain OCT interferometer. Explain depth selection and the formation of A- and B-scans. State one medical application and one limitation.

Expected answer and working. A broadband, low-temporal-coherence light source is split into sample and reference arms. Returning sample backscatter and reference-mirror light recombine at a detector. Significant interference envelope occurs when their optical path lengths match within the coherence length. Moving the reference mirror samples optical delay and gives a depth reflectivity profile (A-scan); lateral translation yields successive A-scans forming a cross-sectional B-scan. Retinal-layer imaging is an application. Tissue scattering/absorption limits penetration, so OCT is best for transparent or superficial structures.



Proposed allocation: Source, splitter, two arms, mirror and detector sketch (2); Optical path matching / coherence gate (1); A-scan and B-scan formation (1); Appropriate application (1); Penetration limitation (1). **Total 6.**

Source map: Syllabus p.1; A2 PDF pp.3-7. **Closest past question:** 2024 Q4(b); 2025 Q4(b). **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.